

ENVIRONMENTAL & SAFETY MONITORING NODE: HARDWARE SETUP MANUAL

1. Project Bill of Materials (BOM)

The following components are utilized in the construction of the stationary environmental checkpoint monitor.

Item	Component	Specification	Qty	Target Function
1	Microcontroller	ESP32 DevKit V1 (30-pin)	1	Processes edge data, executes ISR, handles Wi-Fi payload transmission.
2	Environmental Sensor	DHT22	1	Captures ambient temperature (°C) and relative humidity (%).
3	Emergency Switch	6x6mm Tactile Push Button	1	Triggers the high-priority OS emergency daemon alert.
4	Prototyping Board	MB102 Solderless Breadboard	1	Provides the physical platform to mount and wire all components.
5	Interconnects	Solid-core Jumper Wires (M-M)	6	Links power, ground, and logic lines between devices.

2. Definitive Pin Mapping Table

To prevent shorts and ensure the hardware interacts flawlessly with the compiled firmware binaries, components must be wired exactly according to the mapping table below:

From Component	Component Pin	To ESP32 Pin	Logic Type
DHT22 Sensor	VCC (Pin 1)	3.3V	Power
	DATA (Pin 2)	GPIO 4	Digital Input
	GND (Pin 3)	GND	Ground
Tactile Button	Terminal 1A	GPIO 15	Digital Input
	Terminal 1B	GND	Ground